Ericsson PE Node ELEC-PE-06 Optical Transport Failure Standard Operating Procedure (SOP)

Node: ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_7562

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Ericsson PE Node ELEC-PE-06 at Electronic City Hub is a critical failure domain that necessitates immediate attention. This failure occurs due to the degradation of SFP/XFPs or fiber cuts, resulting in a significant reduction in received (Rx) and transmitted (Tx) power levels. The Ericsson PE Node utilizes Optical Transport Network (OTN) technology, which relies on the accurate transmission and reception of optical signals through SFP/XFPs. A single SFP/XFP failure or fiber cut can propagate a signal degradation, leading to a reduction in Rx/Tx power levels and ultimately, service disruption. Furthermore, the Ericsson PE Node's Optical Interface Module (OIM) is responsible for managing the optical interface, including SFP/XFP management, optical signal monitoring, and alarm generation. The alarm code ALARM_7562, corresponding to the Optical Interface Alarm, is generated when the Rx/Tx power levels drop below the threshold. It is crucial to promptly identify and resolve the root cause of the Optical Transport Failure on Ericsson PE Node ELEC-PE-06 to prevent service disruption and ensure reliable data transmission.

Pre-requisite Checks

- Verify that the Ericsson PE Node ELEC-PE-06 is accessible and powered on.

- Check the redundant link status to ensure that the secondary link is operational and can take over in case of a primary link failure.

- Dispatch remote hands to Electronic City Hub to assist with physical checks and potential hardware replacements.

- Ensure the Ericsson Node Manager (NM) is running and accessible for monitoring and troubleshooting purposes.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss due to optical signal attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface optical 1/1/1-1, Expected Output: Optical
Interface Status: Interface: 1/1/1-1 SFP/XFP Type: SFP-1000BASE-LX10
Rx Power: -25.6 dBm Tx Power: -3.2 dBm, Description: This command
```

```
displays the status of the optical interface, including the SFP/XFP
type, Rx power level, and Tx power level. This information can be used
to identify potential issues with the SFP/XFP or optical signal
degradation.

Command: show interface optical 1/1/1-1 detail, Expected Output:
Optical Interface Details: Interface: 1/1/1-1 SFP/XFP Vendor: Ericsson
SFP/XFP Serial Number: ABC12345 Rx Power Threshold: -28.8 dBm Tx Power
Threshold: -10.2 dBm, Description: This command provides detailed
information about the optical interface, including the SFP/XFP vendor,
serial number, and power thresholds. This information can be used to
verify the SFP/XFP's specifications and ensure it meets the required
power levels.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace interface optical 1/1/1-1 sfp-xfp Ericsson SFP-1000BASE-LX10 ABC12345, Description: Replace the SFP/XFP with a new one that meets the required specifications and power levels. Ensure the new SFP/XFP is compatible with the Ericsson PE Node and the optical interface.

Step 2: Step: Check and replace the optical fiber, Command: show interface optical 1/1/1-1 fiber status, Description: Check the optical fiber status and replace it if necessary. Ensure the new fiber meets the required specifications and power levels.

Post-Check Verification

```
Command: show interface optical 1/1/1-1, Expected Output: Optical
Interface Status: Interface: 1/1/1-1 SFP/XFP Type: SFP-1000BASE-LX10
Rx Power: -20.6 dBm Tx Power: -5.2 dBm, Description: Verify that the
Rx and Tx power levels have returned to normal after the resolution
steps.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: replace interface optical 1/1/1-1 sfp-xfp Ericsson SFP-1000BASE-LX10 DEF67890, Description: Restore the original SFP/XFP if the replacement SFP/XFP caused the issue.

- Step: Restore the original optical fiber, Command: show interface optical 1/1/1-1 fiber status, Description: Restore the original optical fiber if the replacement fiber caused the issue.